

Chapter 8

Calculus For Semimartingales

In this chapter rules of calculus are given for the most general processes for which stochastic calculus is developed, called semimartingales. A semimartingale is process consisting of a sum of a local martingale and a finite variation process. Integration with respect to semimartingales involves integration with respect to local martingales, and these integrals generalize the Itô integral where integration is done with respect to a Brownian motion. Important concepts, such as compensators and the sharp bracket processes are introduced, and Itô's formula in its general form is given.

8.1 Semimartingales

In stochastic calculus only regular processes are considered. These are either continuous processes, or right-continuous with left limits, or left-continuous with right limits. The regularity of the process implies that it can have at most countably many discontinuities, and all of them are jumps (Chapter 1). The definition of a semimartingale presumes a given filtration and processes which we consider are adapted to it. Following the classical approach, see for example, Metivier (1982), Liptser and Shiriyayev (1989) p.85, a semimartingale, is a local martingale plus a process of finite variation. More precisely,

Definition 8.1 *A regular right-continuous with left limits (càdlàg) adapted process is a semimartingale if it can be represented as a sum of two processes: a local martingale $M(t)$ and a process of finite variation $A(t)$, with $M(0) = A(0) = 0$, and*

$$S(t) = S(0) + M(t) + A(t). \quad (8.1)$$

Example 8.1: (Semimartingales)

1. $S(t) = B^2(t)$, where $B(t)$ is a Brownian motion is a semimartingale. $S(t) = M(t) + t$, where $M(t) = B^2(t) - t$ is a martingale and $A(t) = t$ is a finite variation process.
2. $S(t) = N(t)$, where $N(t)$ is a Poisson process with rate λ , is a semimartingale, as it is a finite variation process.
3. One way to obtain semimartingales from known semimartingales is by applying a twice continuously differentiable (C^2) transformation. If $S(t)$ is a semimartingale and f is a C^2 function, then $f(S(t))$ is also a semimartingale. The decomposition of $f(S(t))$ into martingale part and finite variation part is given by Itô's formula, given later. In this way we can assert that, for example, the geometric Brownian motion $e^{\sigma B(t) + \mu t}$ is a semimartingale.
4. A right-continuous with left limits (càdlàg) deterministic function $f(t)$ is a semimartingale if and only if it is of finite variation. Thus $f(t) = t \sin(1/t)$, $t \in (0, 1]$, $f(0) = 0$ is continuous, but not a semimartingale (see Example 1.7).
5. A diffusion, that is, a solution to a stochastic differential equation with respect to Brownian motion, is a semimartingale. Indeed, the Itô integral with respect to $dB(t)$ is a local martingale and the integral with respect to dt is a process of finite variation.
6. Although the class of semimartingales is rather large, there are processes which are not semimartingales. Examples are: $|B(t)|^\alpha$, $0 < \alpha < 1$, where $B(t)$ is the one-dimensional Brownian motion; $\int_0^t (t-s)^{-\alpha} dB(s)$, $0 < \alpha < 1/2$. It requires analysis to show that the above processes are not semimartingales.

For a semimartingale X , the process of jumps ΔX is defined by

$$\Delta X(t) = X(t) - X(t-), \quad (8.2)$$

and represents the jump at point t . If X is continuous, then of course, $\Delta X = 0$.

8.2 Predictable Processes

In this section we describe the class of predictable processes. This class of processes has a central role in the theory. In particular, only predictable processes can be integrated with respect to a semimartingale. Recall that in discrete time a process H is predictable if H_n is $\mathcal{F}_{n-1}$ measurable, that is, H is known with certainty at time n on the basis of information up to time $n-1$. Predictability in continuous time is harder to define. We recall some general definitions of processes starting with the class of adapted processes.

Definition 8.2 A process X is called adapted to filtration $\mathbb{F} = (\mathcal{F}_t)$, if for all t , $X(t)$ is $\mathcal{F}_t$ -measurable.

In construction of the stochastic integral $\int_0^t H(u) dS(u)$, processes H and S are taken to be adapted to $\mathbb{F}$. For a general semimartingale S , the requirement that H is adapted is too weak, it fails to assure measurability of some basic constructions. H must be *predictable*. The exact definition of predictable processes involves σ -fields generated on $\mathbb{R}^+ \times \Omega$ and is given later in Section 8.13. Note that left-continuous processes are predictable, in the sense that $H(t) = \lim_{s \uparrow t} H(s) = H(s-)$. So that if the values of the process before t are known, then the value at t is determined by the limit. For our purposes it is enough to describe a subclass of predictable processes which can be defined constructively.

Definition 8.3 *H is predictable if it is one of the following:*

- a) *a left-continuous adapted process, in particular, a continuous adapted process.*
- b) *a limit (almost sure, in probability) of left-continuous adapted processes.*
- c) *a regular right-continuous process such that, for any stopping time τ , H_τ is $\mathcal{F}_{\tau-}$ -measurable, the σ -field generated by the sets $A \cap \{T < t\}$, where $A \in \mathcal{F}_t$.*
- d) *a Borel-measurable function of a predictable process.*

Example 8.2: Poisson process $N(t)$ is right-continuous and is obviously adapted to its natural filtration. It can be shown, see Example 8.31, that it is not predictable. Its left-continuous modification $N(t-) = \lim_{s \uparrow t} N(s)$ is predictable, because it is adapted and left-continuous by a). Any measurable function (even right-continuous) of $N(t-)$ is also predictable by d).

Example 8.3: Right-continuous adapted processes may not be predictable, even though they can be approached by left-continuous processes, for example, $X_\epsilon(t) = \lim_{\epsilon \rightarrow 0} X((t + \epsilon)-)$.

Example 8.4: Let T be a stopping time. This means that for any t , the set $\{T > t\} \in \mathcal{F}_t$. Consider the process $X(t) = I_{[0, T]}(t)$. It is adapted, because its values are determined by the set $\{T \leq t\}$ ($X(t) = 1$ if and only if $\omega \in \{T \leq t\}$), and $\{T \leq t\} = \{T > t\}^c \in \mathcal{F}_t$. $X(t)$ is also left-continuous. Thus it is a predictable process by a). We also see that T is a stopping time if and only if the process $X(t) = I_{[0, T]}(t)$ is adapted.

Example 8.5: It will be seen later that when filtration is generated by Brownian motion, then any right-continuous adapted process is predictable. This is why in the definition of the Itô integral right-continuous functions are allowed as integrands.

8.3 Doob-Meyer Decomposition

Recall that a process is a submartingale if for all $s < t$, $E(X(t)|\mathcal{F}_s) \geq X(s)$ almost surely

Theorem 8.4 *If X is a submartingale or a local submartingale, then there exists a local martingale $M(t)$ and a unique increasing predictable process $A(t)$, locally integrable, such that*

$$X(t) = X(0) + M(t) + A(t). \quad (8.3)$$

If $X(t)$ is a submartingale of Dirichlet class (D) (see Definition 7.25), then the process A is integrable, that is, $\sup_t EA(t) < \infty$, and $M(t)$ is a uniformly integrable martingale.

Example 8.6:

1. Let $X(t) = B^2(t)$ on a finite interval $t \leq T$. $X(t)$ is a submartingale. Decomposition (8.3) holds with $M(t) = B^2(t) - t$ and $A(t) = t$. Since the interval is finite, M is uniformly integrable and A is integrable.
2. Let $X(t) = B^2(t)$ on the infinite interval $t \geq 0$. Then (8.3) holds with $M(t) = B^2(t) - t$ and $A(t) = t$. Since the interval is infinite M is a martingale, and A is locally integrable; for example take the localizing sequence $\tau_n = n$.
3. Let $X(t) = N(t)$ be a Poisson process with intensity λ . Then X is a submartingale. Decomposition (8.3) holds with $M(t) = N(t) - \lambda t$ and $A(t) = \lambda t$.

Proof of Theorem 8.4 can be found in Rogers and Williams (1990), p.372-375, Dellacherie (1972), Meyer (1966).

Doob's Decomposition

For processes in discrete time, decomposition (8.3) is due to Doob, and it is simple to obtain. Indeed, if X_n is a submartingale, then clearly,

$X_{n+1} = X_0 + \sum_{i=0}^n (X_{i+1} - X_i)$. By adding and subtracting $E(X_{i+1}|\mathcal{F}_i)$, we obtain the Doob decomposition

$$X_{n+1} = X_0 + \sum_{i=0}^n ((X_{i+1} - E(X_{i+1}|\mathcal{F}_i)) + \sum_{i=0}^n (E(X_{i+1}|\mathcal{F}_i) - X_i), \quad (8.4)$$

where martingale and an increasing process are given by

$$M_{n+1} = \sum_{i=0}^n (X_{i+1} - E(X_{i+1}|\mathcal{F}_i)) \quad \text{and} \quad A_{n+1} = \sum_{i=0}^n (E(X_{i+1}|\mathcal{F}_i) - X_i) \quad (8.5)$$

A_n is increasing due to the submartingale property, $E(X_{i+1}|\mathcal{F}_i) - X_i \geq 0$ for all i . It is also predictable, because $E(X_{n+1}|\mathcal{F}_n)$ and all other terms are $\mathcal{F}_n$ measurable.

It is much harder to prove decomposition (8.3) in continuous time and this was done by Meyer.

8.4 Integrals with respect to Semimartingales

In this section the stochastic integral $\int_0^T H(t) dS(t)$ is defined, where $S(t)$ is a semimartingale. Due to representation $S(t) = S(0) + M(t) + A(t)$ the integral with respect to $S(t)$ is the sum of two integrals one with respect to a local martingale $M(t)$ and the other with respect to a finite variation process $A(t)$. The integral with respect to $A(t)$ can be done path by path as the Stieltjes integral, since $A(t)$, although random, is of finite variation.

The integral with respect to the martingale $M(t)$ is new, it is the stochastic integral $\int_0^T H(t) dM(t)$. When $M(t)$ is Brownian motion $B(t)$, it is the Itô integral, defined in Chapter 4. But now martingales are allowed to have jumps and this makes the theory more complicated. The key property used in the definition of the Itô integral is that on finite intervals Brownian motion is a square integrable martingale. This property in its local form plays an important role in the general case. Conditions for the existence of the integral with respect to a martingale involves the martingale's quadratic variation, which was introduced in Section 7.6.

Stochastic Integral with respect to Martingales

For a simple predictable process $H(t)$, given by

$$H(t) = H(0)I_0 + \sum_{i=0}^{n-1} H_i I_{(T_i, T_{i+1}]}(t), \quad (8.6)$$

where $0 = T_0 \leq T_1 \leq \dots \leq T_n \leq T$ are stopping times and H_i 's are $\mathcal{F}_{T_i}$ -measurable, the stochastic integral is defined as the sum

$$\int_0^T H(t) dS(t) = \sum_{i=0}^{n-1} H_i (M(T_{i+1}) - M(T_i)). \quad (8.7)$$

If $M(t)$ is a locally square integrable martingale, then by the L^2 theory (Hilbert space theory) one can extend the stochastic integral from simple predictable processes to the class of predictable processes H such that

$$\sqrt{\int_0^T H^2(t) d[M, M](t)} \quad \text{is locally integrable.} \quad (8.8)$$

If $M(t)$ is a *continuous* local martingale, then the stochastic integral is defined for a wider class of predictable processes H satisfying

$$\int_0^T H^2(t) d[M, M](t) < \infty \quad \text{a.s.} \quad (8.9)$$

Properties of Stochastic Integrals with respect to Martingales

1. Local martingale property. If $M(t)$ is a local martingale, the integral $\int_0^t H(s)dM(s)$ is a local martingale.
2. Isometry property. If $M(t)$ is a square integrable martingale, and H satisfies

$$\mathbb{E}\left(\int_0^T H^2(s)d[M, M](s)\right) < \infty, \quad (8.10)$$

then $\int_0^t H(s)dM(s)$ is a square integrable martingale with zero mean and variance

$$\mathbb{E}\left(\int_0^t H(s)dM(s)\right)^2 = \mathbb{E}\left(\int_0^t H^2(s)d[M, M](s)\right). \quad (8.11)$$

3. If a local martingale $M(t)$ is of finite variation, then the stochastic integral is indistinguishable from the Stieltjes integral.

Example 8.7: Consider Itô integrals with respect to Brownian motion. Since $B(t)$ is a square integrable martingale on $[0, T]$, with $[B, B](t) = t$, we recover that for a predictable H , such that $\mathbb{E} \int_0^T H^2(s)ds < \infty$, $\int_0^t H(s)dB(s)$ is a square integrable martingale with zero mean and variance $\int_0^t \mathbb{E}H^2(s)ds$.

Stochastic Integrals with respect to Semimartingales

Let S be a semimartingale with representation

$$S(t) = S(0) + M(t) + A(t), \quad (8.12)$$

where M is a local martingale and A is a finite variation process. Let H be a predictable process such that conditions (8.13) and (8.8) hold.

$$\int_0^T |H(t)|dV_A(t) < \infty, \quad (8.13)$$

where $V_A(t)$ is the variation process of A . Then the stochastic integral is defined as the sum of integrals,

$$\int_0^t H(t)dS(t) = \int_0^t H(t)dM(t) + \int_0^t H(t)dA(t). \quad (8.14)$$

Since a representation of a semimartingale (8.12) is not unique, one should check that the stochastic integral does not depend on the representation used. Indeed, if $S(t) = S(0) + M_1(t) + A_1(t)$ is another representation, then

$(M - M_1)(t) = -(A - A_1)(t)$. So that $M - M_1$ is a local martingale of finite variation. But for such martingales stochastic and Stieltjes integrals are the same, and it follows that $\int H(t)dM_1(t) + \int H(t)dA_1(t) = \int H(t)dM(t) + \int H(t)dA(t) = \int H(t)dS(t)$.

Since the integral with respect to a local martingale is a local martingale, and the integral with respect to a finite variation process is a process of finite variation, it follows that a stochastic integral with respect to a semimartingale is a semimartingale.

For details see Liptser and Shirayev (1989), p.90-116.

Example 8.8: Let $N(t)$ be a Poisson process. $N(t)$ is of finite variation and the integral $\int_0^T N(t)dN(t)$ is well defined as a Stieltjes integral, $\int_0^T N(t)dN(t) = \sum_{\tau_i \leq T} N(\tau_i)$, where τ_i 's are the jumps of $N(t)$. However, $\int_0^T N(t)dN(t)$ is not the stochastic integral, since $N(t)$ is not predictable, but $\int_0^T N(t-)dN(t)$ is. It is indistinguishable from the integral in the sense of Stieltjes $\int_0^T N(t-)dN(t) = \sum_{\tau_i \leq T} N(\tau_{i-1})$.

Properties of Stochastic Integrals with respect to Semimartingales

Let X be a semimartingale and H a predictable processes, such that the stochastic integral exists for $0 \leq t \leq T$, and denote

$$(H \cdot X)(t) := \int_0^t H(s)dX(s).$$

Then the stochastic integral $H \cdot X$ has the following properties

1. The jumps of the integral occur at the points of jumps of X , and $\Delta(H \cdot X)(t) = H(t)\Delta X(t)$. In particular, a stochastic integral with respect to a continuous semimartingale is continuous.
2. If τ is a stopping time, then the stopped integral is the integral with respect to the stopped semimartingale,

$$\int_0^{t \wedge \tau} H(s)dX(s) = \int_0^t H(s)I(s \leq \tau)dX(s) = \int_0^t H(s)dX(s \wedge \tau).$$

3. If X is of finite variation, then $\int_0^t H(s)dX(s)$ is indistinguishable from the Stieltjes integral computed path by path.
4. Associativity. If $Y(t) = \int_0^t H(s)dX(s)$ is a semimartingale, and if K is a predictable process, such that $(K \cdot Y)(t) = \int_0^t K(s)dY(s)$ is defined, then $K \cdot Y = K \cdot (H \cdot X) = (KH) \cdot X$, that is,

$$\int_0^t K(s)dY(s) = \int_0^t K(s)H(s)dX(s).$$

8.5 Quadratic Variation and Covariation

If X, Y are semimartingales on the common space, then the quadratic covariation process, also known as the *the square bracket* process and denoted $[X, Y](t)$, is defined, as usual, by

$$[X, Y](t) = \lim \sum_{i=0}^{n-1} (X(t_{i+1}^n) - X(t_i^n))(Y(t_{i+1}^n) - Y(t_i^n)), \quad (8.15)$$

where the limit is taken over shrinking partitions $\{t_i^n\}_{i=0}^n$ of the interval $[0, t]$ when $\delta_n = \max_i(t_{i+1}^n - t_i^n) \rightarrow 0$ and is in probability. Taking $Y = X$ we obtain the quadratic variation process of X .

Example 8.9: We have seen that quadratic variation of Brownian motion $B(t)$ is $[B, B](t) = t$ and of Poisson process $N(t)$ is $[N, N](t) = N(t)$.

Properties of Quadratic Variation

We give the fundamental properties of the quadratic variation process with some explanations, but omit the proofs.

1. If X is a semimartingale, then $[X, X]$ exists and is an adapted process.
2. It is clear from the definition that quadratic variation over non-overlapping intervals is the sum of the quadratic variation over each interval. As such, $[X, X](t)$ is non-decreasing function of t . Consequently $[X, X](t)$ is a function of finite variation.
3. It follows from the definition (8.15) that $[X, Y]$ is bilinear and symmetric, that is, $[X, Y] = [Y, X]$ and

$$[\alpha X + Y, \beta U + V] = \alpha\beta[X, U] + \alpha[X, V] + \beta[Y, U] + [Y, V]. \quad (8.16)$$

4. Polarization identity.

$$[X, Y] = \frac{1}{2}([X + Y, X + Y] - [X, X] - [Y, Y]). \quad (8.17)$$

This property follows directly from the previous one.

5. $[X, Y](t)$ is a regular right-continuous (càdlàg) function of finite variation. This follows from the polarization identity, as $[X, Y]$ is the difference of two increasing functions.
6. The jumps of the quadratic covariation process occur only at points where *both* processes have jumps,

$$\Delta[X, Y](t) = \Delta X(t)\Delta Y(t). \quad (8.18)$$

7. If one of the processes X or Y is of finite variation, then

$$[X, Y](t) = \sum_{s \leq t} \Delta X(s) \Delta Y(s). \quad (8.19)$$

Notice that although the summation is taken over all s not exceeding t , there are at most countably many terms different from zero.

The following property is frequently used, it follows directly from (8.19).

Corollary 8.5 *If $X(t)$ is a continuous semimartingale of finite variation, then it has zero quadratic covariation with any other semimartingale $Y(t)$.*

Quadratic Variation of Stochastic Integrals

The quadratic covariation of stochastic integrals has the following property

$$\left[\int_0^\cdot H(s) dX(s), \int_0^\cdot K(s) dY(s) \right](t) = \int_0^t H(s) K(s) d[X, Y](s). \quad (8.20)$$

In particular the quadratic variation of a stochastic integral is given by

$$\left[\int_0^\cdot H(s) dX(s), \int_0^\cdot H(s) dX(s) \right](t) = \int_0^t H^2(s) d[X, X](s), \quad (8.21)$$

and

$$\begin{aligned} \left[\int_0^\cdot H(s) dX(s), Y \right](t) &= \left[\int_0^\cdot H(s) dX(s), \int_0^\cdot 1 dY(s) \right](t) \\ &= \int_0^t H(s) d[X, Y](s). \end{aligned} \quad (8.22)$$

Quadratic variation has a representation in terms of the stochastic integral.

Theorem 8.6 *Let X be a semimartingale null at zero. Then*

$$[X, X](t) = X^2(t) - 2 \int_0^t X(s-) dX(s). \quad (8.23)$$

For a partition $\{t_i^n\}$ of $[0, t]$, consider

$$v_n(t) = \sum_{i=0}^{n-1} (X(t_{i+1}^n) - X(t_i^n))^2.$$

For any fixed t the sequence $v_n(t)$ converges in probability to a limit $[X, X](t)$ as $\max_{i \leq n} (t_{i+1}^n - t_i^n) \rightarrow 0$. Moreover, there is a subsequence n_k , such that the processes $v_{n_k}(t)$, converge uniformly on any bounded time interval to the process $X^2(t) - 2 \int_0^t X(s-) dX(s)$.

We justify (8.23) heuristically. By opening the brackets in the sum for $v_n(t)$, adding and subtracting $X^2(t_i)$, we obtain

$$v_n(t) = \sum_{i=0}^{n-1} \left(X^2(t_{i+1}^n) - X^2(t_i^n) \right) - 2 \sum_{i=0}^{n-1} X(t_i^n) \left(X(t_{i+1}^n) - X(t_i^n) \right).$$

The first sum is $X^2(t)$. The second has for its limit in probability $\int_0^t X(s-)dX(s)$. This argument can be made into a rigorous proof, see for example, Metivier (1982), p.175. Alternatively (8.23) can be established by using Itô's formula.

Using the polarization identity, it is easy to see

Corollary 8.7 *For semimartingales X, Y the quadratic covariation process is given by*

$$[X, Y](t) = X(t)Y(t) - X(0)Y(0) - \int_0^t X(s-)dY(s) - \int_0^t Y(s-)dX(s). \quad (8.24)$$

This is also known as the integration by parts or product rule formula.

8.6 Itô's Formula for Continuous Semimartingales

If $X(t)$ is a *continuous* semimartingale and f is a twice continuously differentiable function, then $Y(t) = f(X(t))$ is a semimartingale and admits the following representation

$$f(X(t)) - f(X(0)) = \int_0^t f'(X(s))dX(s) + \frac{1}{2} \int_0^t f''(X(s))d[X, X](s). \quad (8.25)$$

In differential form this is written as

$$df(X(t)) = f'(X(t))dX(t) + \frac{1}{2}f''(X(t))d[X, X](t). \quad (8.26)$$

It follows, in particular, that $f(X(t))$ is also a semimartingale, and its decomposition into the martingale part and the finite variation part can be obtained from Itô's formula by splitting the stochastic integral with respect to $X(t)$ into the integral with respect to a local martingale $M(t)$ and a finite variation process $A(t)$.

We have given a justification of Itô's formula and examples of its use in Chapters 4 and 6.

Remark 8.1:

1. The differentiability properties of f may be relaxed. If, for example, X is of finite variation, then f needs to be only once continuously differentiable. f can be defined only on an open set, rather than a whole line, but then X must take its values almost surely in this set. For example, if X is a positive semimartingale, then Itô's formula can be used with $f = \ln$.
2. Itô's formula holds for convex functions (Protter (1992) p.163), and more generally, for functions which are the difference of two convex functions. This is the Meyer-Itô (Itô-Tanaka) formula, see for example, Protter (1992) p.167, Rogers and Williams (1990), p.105, Revuz and Yor p.208. In particular, if f is a convex function on $\mathbb{R}$ and $X(t)$ is a semimartingale, then $f(X(t))$ is also a semimartingale. See also Section 8.7.
3. It follows from Itô's formula that if a semimartingale X is continuous with nil quadratic variation $[X, X](t) = 0$, then the differentiation rule is the same as in the ordinary calculus. If $X(t)$ is a Brownian motion, then $d[X, X](t) = dt$ and we recover formulae (4.39) and (4.53). If X has jumps, then the formula has an extra term (see Section 8.10).

The following result is a direct corollary to Itô's formula.

Corollary 8.8 *Let $X(t)$ be a continuous semimartingale and f be twice continuously differentiable. Then*

$$[f(X), f(X)](t) = \int_0^t \left(f'(X(s)) \right)^2 d[X, X](s). \quad (8.27)$$

PROOF: Since $[X, X]$ is of finite variation it follows from (8.25)

$$[f(X), f(X)](t) = \left[\int_0^\cdot f'(X(s)) dX(s), \int_0^\cdot f'(X(s)) dX(s) \right] (t).$$

□

Itô's Formula for Functions of Several Variables

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 , and let $\mathbf{X}(t) = (X_1(t), \dots, X_n(t))$ be a *continuous* semimartingale in $\mathbb{R}^n$, that is, each X_i is a continuous semimartingale. Then $f(\mathbf{X})$ is a semimartingale, and has the following representation.

$$\begin{aligned} f(\mathbf{X}(t)) - f(\mathbf{X}(0)) &= \int_0^t \sum_{i=1}^n \frac{\partial f}{\partial x_i}(\mathbf{X}(s)) dX_i(s) \\ &+ \frac{1}{2} \int_0^t \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{X}(s)) d[X_i, X_j](s). \end{aligned} \quad (8.28)$$

8.7 Local Times

Let $X(t)$ be a continuous semimartingale. Consider $|X(t) - a|$, $a \in \mathbb{R}$. The function $|x - a|$ is not differentiable at a , but at any other point its derivative is given by $\text{sign}(x - a)$, where $\text{sign}(x) = 1$ for $x > 0$ and $\text{sign}(x) = -1$ for $x \leq 0$. It is possible to extend Itô's formula for this case and prove (see Rogers and Williams (1990), p.95-102, Protter (1992) p.165-167)

Theorem 8.9 (Tanaka's Formula) *Let $X(t)$ be a continuous semimartingale. Then for any $a \in \mathbb{R}$ there exists a continuous non-decreasing adapted process $L^a(t)$, called the local time at a of X , such that*

$$|X(t) - a| = |X(0) - a| + \int_0^t \text{sign}(X(s) - a) dX(s) + L^a(t). \quad (8.29)$$

As a function in a , $L^a(t)$ is right-continuous with left limits. For any fixed a as a function in t $L^a(t)$ increases only when $X(t) = a$, that is, $L^a(t) = \int_0^t I(X(s) = a) dL^a(s)$. Moreover, if $X(t)$ is a continuous local martingale, then $L^a(t)$ is jointly continuous in a and t .

Remark 8.2: Heuristically Tanaka's formula can be justified by a formal application of Itô's formula to the function $\text{sign}(x)$. The derivative of $\text{sign}(x)$ is zero everywhere but at zero, where it is not defined. However, it is possible to define the derivative as a generalized function or a Schwartz distribution, in which case it is equal to 2δ . Thus the second derivative of $|x - a|$ is $\delta(x - a)$ in the generalized function sense. The local time at a of X is defined as $L^a(t) = \int_0^t \delta(X(s) - a) ds$. Formal use of Itô's formula gives 8.29.

Theorem 8.10 (Occupation Times Formula) *Let $X(t)$ be a continuous semimartingale with local time $L^a(t)$. Then for any bounded measurable function $g(x)$*

$$\int_0^t g(X(s)) d[X, X](s) = \int_{-\infty}^{\infty} g(a) L^a(t) da. \quad (8.30)$$

In particular

$$[X, X](t) = \int_{-\infty}^{\infty} L^a(t) da. \quad (8.31)$$

Example 8.10: Let $X(t) = B(t)$ be Brownian motion. Then its local time at zero process, $L^0(t)$ satisfies (Tanaka's formula)

$$L^0(t) = |B(t)| - \int_0^t \text{sign}(B(s)) dB(s). \quad (8.32)$$

The occupation times formula (8.30) becomes

$$\int_0^t g(B(s))ds = \int_{-\infty}^{\infty} g(a)L^a(t)da. \quad (8.33)$$

The time Brownian motion spends in a set $A \subset \mathbb{R}$ up to time t is given by (with $g(x) = I_A(x)$)

$$\int_0^t I_A(B(s))ds = \int_{-\infty}^{\infty} I_A(a)L^a(t)da = \int_A L^a(t)da. \quad (8.34)$$

Remark 8.3: Taking $A = (a, a + da)$ and $g(x) = I_{(a, a+da)}(x)$ its indicator in (8.33), $L^a(t)da$ is the time Brownian motion spends in $(a, a + da)$ up to time t , which explains the name “local time”. The time Brownian motion spends in a set A is $\int_A L^a(t)da$, therefore the name “occupation times density” formula (8.34). For a continuous semimartingale the formula (8.30) is the “occupation times density” formula relative to the random “clock” $d[X, X](s)$.

Example 8.11: $X(t) = |B(t)|$ is a semimartingale, since $|x|$ is a convex function. Its decomposition into the martingale and finite variation parts is given by Tanaka’s formula (8.32). $\sqrt{|B(t)|}$ is not a semimartingale, see Protter (1992), p.169-170.

Example 8.12: The function $(x - a)^+$ is important in financial application, as it gives the payoff of a financial stock option. The Meyer-Tanaka’s formula for $(x - a)^+$

$$(X(t) - a)^+ = (X(0) - a)^+ + \int_0^t I(X(s) > a)dX(s) + \frac{1}{2}L_t^a. \quad (8.35)$$

Theorem 8.11 *Let $L^a(t)$ be the local time of Brownian motion at a , and $f_t(a)$ the density of $N(0, t)$ at a . Then*

$$\mathbb{E}(L^a(t)) = \int_0^t f_s(a)ds \quad \text{hence} \quad \frac{d\mathbb{E}(L^a(t))}{dt} = f_t(a). \quad (8.36)$$

PROOF: Taking expectation in both sides of equation (8.33) and changing the order of integration, we obtain for any positive and bounded g

$$\int_{-\infty}^{\infty} g(a)f_s(a)dsda = \int_{-\infty}^{\infty} g(a)\mathbb{E}(L^a(t))da.$$

The result follows, since g is arbitrary. □

A similar result can be established for continuous semimartingales by using equation (8.30) (e.g. Klebaner (2002)).

Remark 8.4: Local times can also be defined for discontinuous semimartingales. For any fixed a , $L^a(t)$ is a continuous non-decreasing function in t , and it increases only at points of continuity of X where it is equal to a , that is, $X(t-) = X(t) = a$. The formula (8.30) holds with quadratic variation $[X, X]$ replaced by its continuous part $[X, X]^c$, see for example, Protter (1992), p.168.

8.8 Stochastic Exponential

The stochastic exponential (also known as the semimartingale, or Doléans-Dade exponential) is a stochastic analogue of the exponential function. Recall that if $f(t)$ is a smooth function then $g(t) = e^{f(t)}$ is the solution to the differential equation $dg(t) = g(t)df(t)$. The stochastic exponential is defined as a solution to a similar stochastic equation. The stochastic exponential of Itô processes was introduced in Section 5.2. For a semimartingale X , its stochastic exponential $\mathcal{E}(X)(t) = U(t)$ is defined as the unique solution to the equation

$$U(t) = 1 + \int_0^t U(s-)dX(s) \quad \text{or} \quad dU(t) = U(t-)dX(t); \quad \text{with } U(0) = 1. \quad (8.37)$$

As an application of Itô's formula and the rules of stochastic calculus we prove

Theorem 8.12 *Let X be a continuous semimartingale. Then its stochastic exponential is given by*

$$U(t) = \mathcal{E}(X)(t) = e^{X(t) - X(0) - \frac{1}{2}[X, X](t)}. \quad (8.38)$$

PROOF: Write $U(t) = e^{V(t)}$, with $V(t) = X(t) - X(0) - \frac{1}{2}[X, X](t)$. Then

$$dU(t) = d(e^{V(t)}) = e^{V(t)}dV(t) + \frac{1}{2}e^{V(t)}d[V, V](t).$$

Using the fact that $[X, X](t)$ is a continuous process of finite variation, we obtain $[X, [X, X]](t) = 0$, and $[V, V](t) = [X, X](t)$. Using this, we obtain

$$dU(t) = e^{V(t)}dX(t) - \frac{1}{2}e^{V(t)}d[X, X](t) + \frac{1}{2}e^{V(t)}d[X, X](t) = e^{V(t)}dX(t),$$

or $dU(t) = U(t)dX(t)$. Thus $U(t)$ defined by (8.38) satisfies (8.37). To show uniqueness, let $V(t)$ be another solution to (8.37), and consider $V(t)/U(t)$. By integration by parts

$$d\left(\frac{V(t)}{U(t)}\right) = V(t)d\left(\frac{1}{U(t)}\right) + \frac{1}{U(t)}dV(t) + d\left[V, \frac{1}{U}\right](t).$$

By Itô's formula, using that $U(t)$ is continuous and satisfies (8.37)

$$d\left(\frac{1}{U(t)}\right) = -\frac{1}{U(t)}dX(t) + \frac{1}{U(t)}d[X, X](t),$$

which leads to

$$d\left(\frac{V(t)}{U(t)}\right) = -\frac{V(t)}{U(t)}dX(t) + \frac{V(t)}{U(t)}dX(t) + \frac{V(t)}{U(t)}d[X, X](t) - \frac{V(t)}{U(t)}d[X, X](t) = 0.$$

Thus $V(t)/U(t) = \text{const.} = V(0)/U(0) = 1$. □

Properties of the stochastic exponential are given by the following result.

Theorem 8.13 *Let X and Y be semimartingales on the same space. Then*

1. $\mathcal{E}(X)(t)\mathcal{E}(Y)(t) = \mathcal{E}(X + Y + [X, Y])(t)$
2. *If X is continuous, $X(0) = 0$, then $(\mathcal{E}(X)(t))^{-1} = \mathcal{E}(-X + [X, X])(t)$.*

The proof uses the integration by parts formula and is left as an exercise.

Example 8.13: (Stock process and its Return process.)

An application in finance is provided by the relation between the stock process and its return process. The return is defined by $dR(t) = dS(t)/S(t-)$. Hence the stock price is the stochastic exponential of the return, $dS(t) = S(t-)dR(t)$, and $S(t) = S(0)\mathcal{E}(R)(t)$.

Stochastic Exponential of Martingales

Stochastic exponential $U = \mathcal{E}(M)$ of a martingale, or a local martingale, $M(t)$ is a stochastic integral with respect to $M(t)$. Since stochastic integrals with respect to martingales or local martingales are local martingales, $\mathcal{E}(M)$ is a local martingale. In applications it is important to have conditions for $\mathcal{E}(M)$ to be a true martingale.

Theorem 8.14 (Martingale exponential) *Let $M(t), 0 \leq t \leq T < \infty$ be a continuous local martingale null at zero. Then its stochastic exponential $\mathcal{E}(M)$ is given by $e^{M(t) - \frac{1}{2}[M, M](t)}$ and it is a continuous positive local martingale. Consequently, it is a supermartingale, it is integrable and has a finite non-increasing expectation. It is a martingale if any of the following conditions hold.*

1. $E(e^{M(T) - \frac{1}{2}[M, M](T)}) = 1$.
2. For all $t \geq 0$, $E\left(\int_0^t e^{2M(s) - [M, M](s)} d[M, M](s)\right) < \infty$.
3. For all $t \geq 0$, $E\left(\int_0^t e^{2M(s)} d[M, M](s)\right) < \infty$.

Moreover, if the expectations above are bounded by $K < \infty$, then $\mathcal{E}(M)$ is a square integrable martingale.

PROOF: By Theorem 8.12, $\mathcal{E}(M)(t) = e^{M(t) - \frac{1}{2}[M, M](t)}$, therefore it is positive. Being a stochastic integral with respect to a martingale, it is a local martingale. Thus $\mathcal{E}(M)$ is a supermartingale, as a positive local martingale, see Theorem 7.23. A supermartingale has a non-increasing expectation, and is a martingale if and only if its expectation at T is the same as at 0 (see Theorem 7.3). This gives the first condition. The second condition follows from Theorem 7.35, which states that if a local martingale has finite expectation of its

quadratic variation, then it is a martingale. So that if $E[\mathcal{E}(M), \mathcal{E}(M)](t) < \infty$, then $\mathcal{E}(M)$ is a martingale. By the quadratic variation of an integral

$$[\mathcal{E}(M), \mathcal{E}(M)](t) = \int_0^t e^{2M(s)-[M,M](s)} d[M, M](s). \quad (8.39)$$

The third condition follows from the second, since $[M, M]$ is positive and increasing. The last statement follows by Theorem 7.35, since the bound implies $\sup_{0 \leq t} E[\mathcal{E}(M), \mathcal{E}(M)](t) < \infty$. □

Theorem 8.15 (Kazamaki's condition) *Let M be a continuous local martingale with $M(0) = 0$. If $e^{\frac{1}{2}M(t)}$ is a submartingale, then $\mathcal{E}(M)$ is a martingale.*

The result is proven in Revuz and Yor (1999), p. 331.

Theorem 8.16 *Let M be a continuous martingale with $M(0) = 0$. If*

$$E\left(e^{\frac{1}{2}M(T)}\right) < \infty, \quad (8.40)$$

then $\mathcal{E}(M)$ is a martingale on $[0, T]$.

PROOF: By Jensen's inequality (see Exercise (7.3)) if g is a convex function, and $E|g(M(t))| < \infty$ for $t \leq T$, then $Eg(M(t)) \leq Eg(M(T))$ and $g(M(t))$ is a submartingale. Since $e^{x/2}$ is convex, the result follows by Kazamaki's condition. □

Theorem 8.17 (Novikov's condition) *Let M be a continuous local martingale with $M(0) = 0$. Suppose that for each $t \leq T$*

$$E\left(e^{\frac{1}{2}[M,M](t)}\right) < \infty. \quad (8.41)$$

Then $\mathcal{E}(M)$ is a martingale with mean one. In particular, if for each t there is a constant K_t such that $[M, M](t) < K_t$, then $\mathcal{E}(M)(t)$, $t \leq T$ is a martingale.

PROOF: The condition (8.41) implies that $[M, M](t)$ has moments. By the BDG inequality (7.41), $\sup_{t \leq T} M(t)$ is integrable, therefore $M(t)$ is a martingale. Next, using the formula for $\mathcal{E}(M)$ and Jensen's inequality, $E\sqrt{X} \leq \sqrt{E(X)}$,

$$E(e^{\frac{1}{2}M(T)}) = E\sqrt{\mathcal{E}(M)(T)e^{\frac{1}{2}[M,M](T)}} \leq \sqrt{E\mathcal{E}(M)(T)E\left(e^{\frac{1}{2}[M,M](T)}\right)} < \infty. \quad (8.42)$$

The last expression is finite since $E\mathcal{E}(M)(T) < \infty$, as $\mathcal{E}(M)$ is a supermartingale, and by the condition of the theorem. This shows that the condition (8.40) holds and the result follows. $\square$

Another proof can be found in Karatzas and Shreve (1988), p.198.

Now we can complete the missing step in the proof of Levy's theorem on characterization of Brownian motion (Theorem 7.36).

Corollary 8.18 *If $M(t)$ is a continuous local martingale with $[M, M](t) = t$, then $U(t) = \mathcal{E}(uM)(t) = e^{uM(t) - u^2 t/2}$ is a martingale.*

PROOF: Clearly $uX(t)$ is a continuous local martingale with quadratic variation $u^2 t$. The result follows by Novikov's condition.

However, it is possible to give a proof from first principles, by using the BDG inequality, and the fact that under the condition (8.10) stochastic integrals with respect to martingales are martingales. Since U is a stochastic exponential it satisfies the SDE

$$U(t) = 1 + \int_0^t U(s) d(uM(s)).$$

A sufficient condition for a stochastic integral to be a martingale is finiteness of the expectation of its quadratic variation. Thus it is enough to check the condition

$$E \int_0^T U^2(t) d[uM, uM](t) = u^2 E \int_0^T U^2(t) dt < \infty.$$

To see this use the BDG inequality with $p = 2$ (for $U(t) - 1$ to have 0 at $t = 0$).

$$E(U^2(T) - 1) \leq E(\sup_{t \leq T} U(t))^2 - 1 \leq CE[U, U](T) = C \int_0^T E(U^2(t)) dt.$$

Let $h(t) = E(U^2(t))$. Then

$$h(T) \leq 1 + C \int_0^T h(t) dt.$$

Gronwall's inequality (Theorem 1.20) implies that $h(T) \leq e^{CT} < \infty$. Thus $E(U^2(T)) < \infty$, and $U(t)$ is a martingale. $\square$

The next section gives more information and tools for processes with jumps.

8.9 Compensators and Sharp Bracket Process

A process N is called increasing if all of its realizations $N(t)$ are non-decreasing functions of t . A process N is of finite variation if all of its realizations $N(t)$ are functions of finite variation, $V_N(t) < \infty$ for all t , where $V_N(t)$ is the variation of N on $[0, t]$.

Definition 8.19 *An increasing process N , $t \geq 0$, is called integrable if $\sup_{t \geq 0} \mathbb{E}N(t) < \infty$.*

A finite variation process N is of integrable variation if its variation process is integrable, $\sup_{t \geq 0} \mathbb{E}V_N(t) < \infty$.

A finite variation process N is of locally integrable variation if there is a sequence of stopping times τ_n such that $\tau_n \uparrow \infty$ so that $N(t \wedge \tau_n)$ is of integrable variation, that is, $\sup_{t \geq 0} \mathbb{E}V_N(t \wedge \tau_n) < \infty$.

Example 8.14: A Poisson process $N(t)$ with parameter λ is of finite but not integrable variation, since for any t , $V_N(t) = N(t) < \infty$, but $\sup_{t \geq 0} \mathbb{E}V_N(t) = \infty$. It is of locally integrable variation, since $\sup_{t \geq 0} \mathbb{E}V_N(t \wedge n) = \lambda n < \infty$. Here $\tau_n = n$.

Example 8.15: It can be seen that a finite variation process $N(t)$ with bounded jumps $|\Delta N(t)| \leq c$ is of locally integrable variation. If $\tau_n = \inf\{t : V_N(t) \geq n\}$, then $N(t \wedge \tau_n)$ has variation bounded by $n + c$. τ_n are stopping times, as first times of boundary crossing.

Definition 8.20 *Let $N(t)$ be an adapted process of integrable or locally integrable variation. Its compensator $A(t)$ is the unique predictable process such that $M(t) = N(t) - A(t)$ is a local martingale.*

Existence of compensators is assured by the Doob-Meyer decomposition.

Theorem 8.21 *Let $N(t)$ be an adapted process of integrable or locally integrable variation. Then its compensator exists. Moreover, it is locally integrable.*

PROOF: As a finite variation process is a difference of two increasing processes, it is enough to establish the result for increasing processes. By localization it is possible to assume that it is integrable. But an increasing integrable process is a submartingale, and the result follows by the Doob-Meyer decomposition Theorem 8.4. □

Remark 8.5: The condition $M = N - A$ is a local martingale is equivalent to the condition (see Liptser and Shirayev (1989), p.33)

$$\mathbb{E} \int_0^\infty H(s) dN(s) = \mathbb{E} \int_0^\infty H(s) dA(s), \quad (8.43)$$

for any positive predictable process H . Sometimes this integral condition (8.43) is taken as the definition of the compensator, e.g. Karr (1986).

The compensator of N is also called the dual predictable projection of N (Rogers and Williams (1990), p.350, Liptser and Shiriyayev (1989), p.33).

Note that the compensator is unique with respect to the given filtration and probability. If the filtration or probability are changed, then the compensator will also change.

Recall that the quadratic variation process $[X, X](t)$ of a semimartingale X exists and is non-decreasing. Consider now semimartingales with integrable ($\sup_{t \geq 0} E[X, X](t) < \infty$) or locally integrable quadratic variation.

Definition 8.22 *The sharp bracket (or angle bracket, or predictable quadratic variation) $\langle X, X \rangle(t)$ process of a semimartingale X is the compensator of $[X, X](t)$. That is, it is the unique predictable process that makes $[X, X](t) - \langle X, X \rangle(t)$ into a local martingale.*

Example 8.16: Let N be a Poisson process. It is of finite variation and changes only by jumps (pure jump process), which are of size 1, $\Delta N(t) = 0$ or 1, and $(\Delta N(t))^2 = \Delta N(t)$. Its quadratic variation is the process $N(t)$ itself,

$$[N, N](t) = \sum_{0 \leq s \leq t} (\Delta N(s))^2 = \sum_{0 \leq s \leq t} \Delta N(s) = N(t).$$

Clearly $\sup_{0 \leq t \leq T} E[N, N](t) = T$. Thus N is of integrable variation on $[0, T]$. t is non-random, therefore predictable. Since $[N, N](t) - t = N(t) - t$ is a martingale,

$$\langle N, N \rangle(t) = t.$$

Example 8.17: Let B be a Brownian motion. Its quadratic variation is $[B, B](t) = t$, and since it is non-random, it is predictable. Hence $\langle B, B \rangle(t) = t$, and the martingale part in the Doob-Meyer decomposition of $[B, B](t)$ is trivial, $M(t) \equiv 0$.

The last example generalizes to any continuous semimartingale.

Theorem 8.23 *If $X(t)$ is a continuous semimartingale with integrable quadratic variation, then $\langle X, X \rangle(t) = [X, X](t)$, and there is no difference between the sharp and the square bracket processes.*

PROOF: The quadratic variation jumps at the points of jumps of X and $\Delta[X, X](s) = (\Delta X(s))^2$. Since X has no jumps, $[X, X](t)$ is continuous. $[X, X](t)$ is predictable as a continuous and adapted process, the martingale part in the Doob-Meyer decomposition of $[X, X](t)$ is trivial, $M(t) \equiv 0$, and $\langle X, X \rangle(t) = [X, X](t)$.

□

Example 8.18: Let X be a diffusion solving the SDE $dX(t) = \mu(X(t))dt + \sigma(X(t))dB(t)$. Then $[X, X](t) = \int_0^t \sigma^2(X(s))ds = \langle X, X \rangle(t)$.

Sharp Bracket for Square Integrable Martingales

Let M be a square integrable martingale, that is, $\sup_t EM^2(t) < \infty$. Recall that the quadratic variation of M has the property

$$M^2(t) - [M, M](t) \text{ is a martingale.} \quad (8.44)$$

M^2 is a submartingale, since x^2 is a convex function. Using the Doob-Meyer decomposition for submartingales, Theorem 8.4, we can prove the following

Theorem 8.24 *Let M be a square integrable martingale. Then the sharp bracket process $\langle M, M \rangle(t)$ is the unique predictable increasing process for which*

$$M^2(t) - \langle M, M \rangle(t) \text{ is a martingale.} \quad (8.45)$$

PROOF: By the definition of the sharp bracket process, $[M, M](t) - \langle M, M \rangle(t)$ is a martingale. As a difference of two martingales, $M^2(t) - \langle M, M \rangle(t)$ is also a martingale. Since $\langle M, M \rangle(t)$ is predictable and $M^2(t)$ is a submartingale, uniqueness follows by the Doob-Meyer decomposition. $\square$

By taking expectations in (8.45) we obtain a useful corollary.

Corollary 8.25 *Let $M(t)$ be a square integrable martingale. Then*

$$E(M^2(T)) = E[M, M](T) = E\langle M, M \rangle(T). \quad (8.46)$$

This result allows us to use Doob's martingale inequality with the sharp bracket

$$E\left(\left(\sup_{s \leq T} M(s)\right)^2\right) \leq 4E(M^2(T)) = 4E(\langle M, M \rangle(T)). \quad (8.47)$$

By using localization in the proof of Theorem 8.24 one can show

Theorem 8.26 *Let M be a locally square integrable martingale, then the predictable quadratic variation $\langle M, M \rangle(t)$ is the unique predictable process for which $M^2(t) - \langle M, M \rangle(t)$ is a local martingale.*

Next result allows to decide when a local martingale is a martingale by using the predictable quadratic variation.

Theorem 8.27 *Let $M(t)$, $0 \leq t \leq T < \infty$, be a local martingale such that for all t , $E\langle M, M \rangle(t) < \infty$. Then M is a square integrable martingale, moreover $EM^2(t) = E[M, M](t) = E\langle M, M \rangle(t)$. If $T = \infty$, and $\sup_{t < \infty} E\langle M, M \rangle(t) < \infty$, then $M(t)$ is a square integrable martingale on $[0, \infty)$.*

PROOF: $[M, M](t) - \langle M, M \rangle(t)$ is a local martingale. Let τ_n be a localizing sequence. Then $E[M, M](t \wedge \tau_n) = E\langle M, M \rangle(t \wedge \tau_n)$. Since both sides are non-decreasing, they converge to the same limit as $n \rightarrow \infty$. But $\lim_{n \rightarrow \infty} E\langle M, M \rangle(t \wedge \tau_n) = E\langle M, M \rangle(t) < \infty$. Therefore $E[M, M](t) = E\langle M, M \rangle(t) < \infty$. Thus the conditions of Theorem 7.35 are satisfied and the result follows. $\square$

Since a continuous local martingale is locally square integrable, we obtain

Corollary 8.28 *The sharp bracket process (predictable quadratic variation) for a continuous local martingale exists.*

Continuous Martingale Component of a Semimartingale

A function of finite variation has a decomposition into continuous and discrete parts. A semimartingale is a sum of a process of finite variation and a local martingale. It turns out that one can decompose any local martingale into a continuous local martingale and a purely discontinuous one. Such decomposition requires a different approach to the case of finite variation processes.

Definition 8.29 *A local martingale is purely discontinuous if it is orthogonal to any continuous local martingale. Local martingales M and N are orthogonal if MN is a local martingale.*

Example 8.19: A compensated Poisson process $\bar{N}(t) = N(t) - t$ is a purely discontinuous martingale. Let $M(t)$ be any continuous local martingale. Then by the integration by parts formula (8.24)

$$M(t)\bar{N}(t) = \int_0^t M(s-)d\bar{N}(s) + \int_0^t \bar{N}(s-)dM(s) + [M, \bar{N}](t).$$

Since $\bar{N}$ is of finite variation, by the property (8.19) of quadratic covariation $[M, \bar{N}](t) = \sum_{s \leq t} \Delta M(s) \Delta \bar{N}(s)$. But M is continuous, $\Delta M(s) = 0$, and $[M, \bar{N}](t) = 0$. Therefore, $M(t)\bar{N}(t)$ is a sum of two stochastic integrals with respect to local martingales, and itself is a local martingale.

It is possible to show that any local martingale M has a unique decomposition

$$M = M^c + M^d,$$

where M^c is a continuous and M^d a purely discontinuous local martingale (see for example Liptser and Shiryaev (1989), p.39, Protter (1992), Jacod and Shiryaev (1987)).

If X is a semimartingale with representation

$$X(t) = X(0) + M(t) + A(t), \tag{8.48}$$

with a local martingale M , then M^c is called the continuous martingale component of X and is denoted by X^{cm} . Even if the above representation of X is not unique, the continuous martingale component of X is the same for all representations. Indeed, if $X(t) = X(0) + M_1(t) + A_1(t)$, is another representation, then $(M - M_1)(t) = -(A - A_1)(t)$. Hence $(M - M_1)$ is a martingale of finite variation. Hence its continuous component is also a martingale of finite variation. But a continuous martingale of finite variation is a constant. This implies that $M^c - M_1^c = 0$. Thus $X^{cm} = M^c$ is the same for all representations. If X is of finite variation then the martingale part is zero and by the uniqueness of X^{cm} we have

Corollary 8.30 *If X is a semimartingale of finite variation, then $X^{cm} \equiv 0$.*

For example, the compensated Poisson process $N(t) - t$ has zero continuous martingale component.

It can be shown that

$$\langle X^{cm}, X^{cm} \rangle = [X, X]^c, \quad (8.49)$$

where $[X, X]^c$ is the continuous part of the finite variation process $[X, X]$. Of course, because X^{cm} is continuous $\langle X^{cm}, X^{cm} \rangle = [X^{cm}, X^{cm}]$.

Let $\Delta X(s) = X(s) - X(s-)$ and put $X(0-) = 0$ and $[X, X]^c(0) = 0$. Since the jumps of quadratic variation satisfy (see (8.18))

$$\Delta[X, X](s) = (\Delta X(s))^2,$$

$$\begin{aligned} [X, X](t) &= [X, X]^c(t) + \sum_{0 < s \leq t} \Delta[X, X](s) = [X, X]^c(t) + \sum_{0 < s \leq t} (\Delta X(s))^2, \\ &= \langle X^{cm}, X^{cm} \rangle + \sum_{0 < s \leq t} (\Delta X(s))^2. \end{aligned} \quad (8.50)$$

Since the quadratic variation $[X, X]$ for a semimartingale exists, and the predictable the quadratic variation $\langle X^{cm}, X^{cm} \rangle$ exists (Corollary 8.28) we obtain

Corollary 8.31 *If X is a semimartingale then for each t*

$$\sum_{s \leq t} (\Delta X(s))^2 < \infty. \quad (8.51)$$

Conditions for Existence of a Stochastic Integral

The class of processes H for which the stochastic integral with respect to a martingale M can be defined depends in an essential way on the properties of the predictable quadratic variation $\langle M, M \rangle$ of M . Consider integrals with

respect to a locally square integrable martingale M , possibly discontinuous. The stochastic integral $\int_0^T H(s) dM(s)$ can be defined for predictable processes H such that

$$\int_0^T H^2(t) d\langle M, M \rangle(t) < \infty, \quad (8.52)$$

and in this case the integral $\int_0^t H(s) dM(s)$, $0 \leq t \leq T$, is a local martingale.

The class of processes H that can be integrated against M is wider when $\langle M, M \rangle(t)$ is continuous, and even wider when $\langle M, M \rangle(t)$ is absolutely continuous (can be represented as an integral with respect to dt). These classes and conditions are given, for example, in Liptser and Shiryaev (2001), p. 191.

Example 8.20: Let filtration $\mathbf{F}$ be generated by a Brownian motion $B(t)$ and a Poisson process $N(t)$. The process $N(t-)$ is a left-continuous modification of $N(t)$. By definition, $N(t-) = \lim_{s \uparrow t} N(s)$. Being left-continuous it is predictable. Condition (8.52) holds. The integral $\int_0^t N(s-) dB(s)$ is a well defined a stochastic integral of a predictable process with respect to a martingale B .

Properties of the Predictable Quadratic Variation

The predictable quadratic variation (the sharp bracket process) has similar properties to the quadratic variation (the square bracket) process. We list them without proof. All the processes below are assumed to be semimartingales with locally integrable quadratic variation.

1. $\langle X, X \rangle(t)$ is increasing in t .
2. $\langle X, Y \rangle$ is bilinear and symmetric,

$$\langle \alpha X + Y, \beta U + V \rangle = \alpha\beta \langle X, U \rangle + \alpha \langle X, V \rangle + \beta \langle Y, U \rangle + \langle Y, V \rangle. \quad (8.53)$$

3. Polarization identity. $\langle X, Y \rangle = \frac{1}{4}(\langle X + Y, X + Y \rangle - \langle X - Y, X - Y \rangle)$.
4. $\langle X, Y \rangle$ is a predictable process of finite variation.
5. $\langle X, Y \rangle = 0$ if X or Y is of finite variation and one of them is continuous.
6. The sharp bracket process of stochastic integrals $\langle H \cdot X, K \cdot Y \rangle(t)$.

$$\left\langle \int_0^\cdot H(s) dX(s), \int_0^\cdot K(s) dY(s) \right\rangle(t) = \int_0^t H(s) K(s) d\langle X, Y \rangle(s), \quad (8.54)$$

in particular $\left\langle \int_0^\cdot H(s) dX(s), \int_0^\cdot H(s) dX(s) \right\rangle(t) = \int_0^t H^2(s) d\langle X, X \rangle(s)$,
 $\left\langle \int_0^\cdot H(s) dX(s), Y \right\rangle(t) = \left\langle \int_0^\cdot H(s) dX, \int_0^\cdot dY \right\rangle(t) = \int_0^t H(s) d\langle X, Y \rangle(s).$

Recall that stochastic integrals with respect to local martingales are again local martingales. Using the sharp bracket of the integral together with Theorem 8.27 we obtain

Theorem 8.32 *Let $M(t)$, $0 \leq t \leq T$ be a local martingale and $H(t)$ be a predictable process such that $\mathbb{E}\left(\int_0^T H^2(s) d\langle M, M \rangle(s)\right) < \infty$. Then $\int_0^t H(s) dM(s)$ is a square integrable martingale, moreover*

$$\left\langle \int_0^\cdot H(s) dM(s), \int_0^\cdot H(s) dM(s) \right\rangle(t) = \int_0^t H^2(s) d\langle M, M \rangle(s). \quad (8.55)$$

Using this result, we obtain the isometry property for stochastic integrals in terms of the sharp bracket process.

$$\mathbb{E}\left(\int_0^t H(s) dM(s)\right)^2 = \mathbb{E}\left(\int_0^t H^2(s) d\langle M, M \rangle(s)\right) \quad (8.56)$$

Example 8.21: Let $M(t)$ be the compensated Poisson process, $M(t) = N(t) - t$, and H be predictable, satisfying $\mathbb{E} \int_0^T H^2(t) dt < \infty$. Then $\int_0^t H(s) dM(s)$ is a martingale, moreover

$$\mathbb{E} \int_0^t H(s) dM(s) = 0, \text{ and } \mathbb{E}\left(\int_0^t H(s) dM(s)\right)^2 = \mathbb{E} \int_0^t H^2(s) ds.$$

8.10 Itô's Formula for Semimartingales

Let $X(t)$ be a semimartingale and f be a C^2 function. Then $f(X(t))$ is a semimartingale, and Itô's formula holds

$$\begin{aligned} f(X(t)) - f(X(0)) &= \int_0^t f'(X(s-)) dX(s) + \frac{1}{2} \int_0^t f''(X(s-)) d[X, X](s) \\ &+ \sum_{s \leq t} \left(f(X(s)) - f(X(s-)) - f'(X(s-)) \Delta X(s) - \frac{1}{2} f''(X(s-)) (\Delta X(s))^2 \right). \end{aligned} \quad (8.57)$$

The quadratic variation $[X, X]$ jumps at the points of jumps of X and its jumps $\Delta[X, X](s) = (\Delta X(s))^2$. Thus the jump part of the integral $\int_0^t f''(X(s-)) d[X, X](s)$ is given by $\sum_{s \leq t} f''(X(s-)) (\Delta X(s))^2$, leading to an equivalent form of the formula

$$\begin{aligned} f(X(t)) - f(X(0)) &= \int_0^t f'(X(s-)) dX(s) + \frac{1}{2} \int_0^t f''(X(s-)) d[X, X]^c(s) \\ &+ \sum_{s \leq t} \left(f(X(s)) - f(X(s-)) - f'(X(s-)) \Delta X(s) \right), \end{aligned} \quad (8.58)$$

where $[X, X]^c$ is the continuous component of the finite variation function $[X, X]$. Using the relationship between the square and the sharp brackets (8.50), we can write Itô's formula with the sharp bracket process of X , provided the sharp bracket exists,

$$\begin{aligned} f(X(t)) - f(X(0)) &= \int_0^t f'(X(s-))dX(s) + \frac{1}{2} \int_0^t f''(X(s-))d\langle X^{cm}, X^{cm} \rangle(s) \\ &\quad + \sum_{s \leq t} \left(f(X(s)) - f(X(s-)) - f'(X(s-))\Delta X(s) \right), \end{aligned} \quad (8.59)$$

where X^{cm} denotes the continuous martingale part of X .

Example 8.22: Let $N(t)$ be a Poisson process. We calculate $\int_0^t N(s-)dN(s)$. The answer can be derived from the integration by parts formula (8.24), but now we use (8.59). Since $(N(t) - t)^{cm} = 0$ (by Corollary 8.30)

$$N^2(t) = 2 \int_0^t N(s-)dN(s) + \sum_{s \leq t} \left(N^2(s) - N^2(s-) - 2N(s-)\Delta N(s) \right).$$

Since $N(s) = N(s-) + \Delta N(s)$, $(N(s-) + \Delta N(s))^2 - N^2(s-) - 2N(s-)\Delta N(s) = (\Delta N(s))^2 = \Delta N(s)$, and the sum simplifies to $\sum_{s \leq t} \Delta N(s) = N(t)$. Thus we obtain

$$\int_0^t N(s-)dN(s) = \frac{1}{2} (N^2(t) - N(t)).$$

A formula (8.59) for a function of n variables reads: $\mathbf{X}(t) = (X^1(t), \dots, X^n(t))$ is a semimartingale and f is a C^2 function of n variables,

$$\begin{aligned} f(\mathbf{X}(t)) - f(\mathbf{X}(0)) &= \sum_{i=1}^n \int_0^t \frac{\partial f}{\partial x_i}(\mathbf{X}(s-))dX^i(s) \\ &\quad + \frac{1}{2} \sum_{i,j=1}^n \int_0^t \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{X}(s-))d\langle X^{i,cm}, X^{j,cm} \rangle(s) \\ &\quad + \sum_{s \leq t} \left(f(\mathbf{X}(s)) - f(\mathbf{X}(s-)) - \sum_{i=1}^n \frac{\partial}{\partial x_i} f(\mathbf{X}(s-))\Delta X^i(s) \right). \end{aligned} \quad (8.60)$$

Itô's formula can be found in many texts, Protter (1992), p. 71, Rogers and Williams (1990), p. 394, Liptser and Shirayev (1989), Métivier (1982), Dellacherie and Meyer (1982).

8.11 Stochastic Exponential and Logarithm

As an application of Itô's formula and the rules of stochastic calculus we outline a proof of the following result.

Theorem 8.33 *Let X be a semimartingale. Then the stochastic equation*

$$U(t) = 1 + \int_0^t U(s-)dX(s). \quad (8.61)$$

has a unique solution, called the stochastic exponential of X , and this solution is given by

$$U(t) = \mathcal{E}(X)(t) = e^{X(t) - X(0) - \frac{1}{2}\langle X, X \rangle^c(t)} \prod_{s \leq t} (1 + \Delta X(s)) e^{-\Delta X(s)}. \quad (8.62)$$

Formula (8.62) can be written by using quadratic variation as follows

$$\mathcal{E}(X)(t) = e^{X(t) - X(0) - \frac{1}{2}[X, X](t)} \prod_{s \leq t} (1 + \Delta X(s)) e^{(-\Delta X(s) + \frac{1}{2}(\Delta X(s))^2)}. \quad (8.63)$$

PROOF: Let $Y(t) = X(t) - X(0) - \frac{1}{2}\langle X, X \rangle^c(t)$ and $V(t) = \prod_{s \leq t} (1 + \Delta X(s)) e^{-\Delta X(s)}$. Note that although the product is taken for all $s \leq t$, there are at most countably many points at which $\Delta X(s) \neq 0$, (by the regularity property of the process), hence there are at most countably many elements different from 1 in the product. We show that the product converges. Since by (8.51) $\sum_{s \leq t} (\Delta X(s))^2 < \infty$, there are only finitely many points s at which $|\Delta X(s)| > 0.5$, which give a finite non-zero contribution to the product. Taking the product with over s at which $|\Delta X(s)| \leq 1/2$, and taking logarithm, it is enough to show that $\sum_{s \leq t} |\ln(1 + \Delta X(s)) - \Delta X(s)|$ converges. But this follows from the inequality $|\ln(1 + \Delta X(s)) - \Delta X(s)| \leq (\Delta X(s))^2$ by (8.51). To see that $U(t)$ defined by (8.62) satisfies (8.61) use Itô's formula applied to the function $f(Y(t), V(t))$ with $f(x_1, x_2) = e^{x_1} x_2$. For the uniqueness of the solution of (8.61) and other details see Liptser and Shirayev (1989), p.123. $\square$

Example 8.23: The stochastic exponential (8.62) of a Poisson process is easily seen to be $\mathcal{E}(N)(t) = 2^{N(t)}$.

If $U = \mathcal{E}(X)$ is the stochastic exponential of X , then $X = \mathcal{L}(U)$ is the stochastic logarithm of U , satisfying equation (8.61)

$$dX(t) = \frac{dU(t)}{U(t-)}, \text{ or } \mathcal{L}(\mathcal{E}(X)) = X.$$

For Itô processes an expression for $X(t)$ is given in Theorem 5.3, for general case see Exercise 8.17.

8.12 Martingale (Predictable) Representations

In this section we give results on representation of martingales by stochastic integrals of predictable processes, also called *predictable* representations. Let $M(t)$ be a martingale, $0 \leq t \leq T$, adapted to the filtration $\mathbb{F} = (\mathcal{F}_t)$, and $H(t)$ be a predictable process satisfying $\int_0^T H^2(s) d\langle M, M \rangle(s) < \infty$ with probability one. Then $\int_0^t H(s) dM(s)$ is a local martingale. The predictable representation property means that the converse is also true. Let $\mathbb{F}^M = (\mathcal{F}_t^M)$ denote the natural filtration of M .

Definition 8.34 *A local martingale M has the predictable representation property if for any $\mathbb{F}^M$ -local martingale X there is a predictable process H such that*

$$X(t) = X(0) + \int_0^t H(s) dM(s). \quad (8.64)$$

This definition is different to the classical one for martingales with jumps, see Remark 8.7 below, but is the same for continuous martingales.

Brownian motion has the predictable representation property (see, for example Revuz and Yor (1999) p. 209, Liptser and Shiryaev (2001) I p. 170).

Theorem 8.35 (Brownian Martingale Representation)

Let $X(t)$, $0 \leq t \leq T$, be a local martingale adapted to the Brownian filtration $\mathbb{F}^B = (\mathcal{F}_t)$. Then there exists a predictable process $H(t)$ such that $\int_0^T H^2(s) ds < \infty$ with probability one, and equation (8.65) holds

$$X(t) = X(0) + \int_0^t H(s) dB(s). \quad (8.65)$$

Moreover, if Y is an integrable $\mathcal{F}_T$ -measurable random variable, $E|Y| < \infty$, then

$$Y = EY + \int_0^T H(t) dB(t). \quad (8.66)$$

If in addition, Y and B have jointly a Gaussian distribution, then the process $H(t)$ in (8.66) is deterministic.

PROOF: We don't prove the representation of a martingale, but only the representation for a random variable based on it.

Take $X(t) = E(Y|\mathcal{F}_t)$. Then $X(t)$, $0 \leq t \leq T$, is a martingale (see Theorem 7.9). Hence by the martingale representation there exists H , such that

$X(t) = X(0) + \int_0^t H(s) dB(s)$. Taking $t = T$ gives the result.

□

Remark 8.6: A functional of the path of the Brownian motion $B_{[0,T]}$ is a random variable Y , $\mathcal{F}_T$ -measurable. Theorem 8.35 states that under the above assumptions, any functional of Brownian motion has the form (8.66).

Since Itô integrals are continuous, and any local martingale of a Brownian filtration is an Itô integral, it follows that all local martingales of a Brownian filtration are continuous. In fact we have the following result (the second statement is not straightforward)

Corollary 8.36

1. All local martingales of the Brownian filtration are continuous.
2. All right-continuous adapted processes are predictable.

Corollary 8.37

Let $X(t)$, $0 \leq t \leq T$, be a square integrable martingale adapted to the Brownian filtration $\mathbf{F}$. Then there exists a predictable process $H(t)$ such that $E \int_0^T H^2(s) ds < \infty$ and representation (8.65) holds. Moreover,

$$\langle X, B \rangle(t) = \int_0^t H(s) ds, \quad \text{and} \quad H(t) = \frac{d\langle X, B \rangle(t)}{dt}. \quad (8.67)$$

The equation (8.67) follows from (8.65) by the rule of the sharp bracket for integrals.

Example 8.24: (Representation of martingales)

1. $X(t) = B^2(t) - t$. Then $X(t) = \int_0^t 2B(s) dB(s)$. Here $H(t) = 2B(t)$, which can also be found by using (8.67).
2. Let $X(t) = f(B(t), t)$ be a martingale. By Itô's formula $dX(t) = \frac{\partial f}{\partial x}(B(t), t) dB(t)$. Thus $H(t) = \frac{\partial f}{\partial x}(B(t), t)$. This also shows that

$$\frac{\langle f(B, t), B \rangle(t)}{dt} = \frac{\partial f}{\partial x}(B(t), t).$$

Example 8.25: (Representation of random variables)

1. If $Y = \int_0^T B(s) ds$, then $Y = \int_0^T (T-s) dB(s)$.
2. $Y = B^2(1)$. Then $M(t) = E(B^2(1) | \mathcal{F}_t) = B^2(t) + (1-t)$. Using Itô's formula for $M(t)$ we obtain $B^2(1) = 1 + 2 \int_0^1 B(t) dB(t)$.

Similar results hold for the Poisson process filtration.

Theorem 8.38 (Poisson Martingale Representation)

Let $M(t)$, $0 \leq t \leq T$, be a local martingale adapted to the Poisson filtration. Then there exists a predictable process $H(t)$ such that

$$M(t) = M(0) + \int_0^t H(s) d\tilde{N}(s), \quad (8.68)$$

where $\tilde{N}(t) = N(t) - t$ is the compensated Poisson process.

When a filtration is larger than the natural filtration of a martingale, then there is the following result (Revuz and Yor (1999) p. 209, Liptser and Shiryaev (2001) p. 170).

Theorem 8.39 *If $M(t)$, $0 \leq t \leq T$, is any continuous local martingale, and X a continuous $\mathbb{F}^M$ -local martingale. Then X has a representation*

$$X(t) = X(0) + \int_0^t H(s) dM(s) + Z(t), \quad (8.69)$$

where H is predictable and $\langle M, Z \rangle = 0$; (consequently $\langle X - Z, Z \rangle = 0$).

Example 8.26: Let $\mathbb{F}$ be generated by two independent Brownian motions B and W , and let $M(t) = \int_0^t W(s) dB(s)$. It is a martingale, as a stochastic integral satisfying $E \int_0^t W^2(s) ds < \infty$. We show that M does not have the predictable representation property.

$\langle M, M \rangle(t) = \int_0^t W^2(s) ds$. Hence $W^2(t) = \frac{d\langle S, S \rangle(t)}{dt}$, which shows that $W^2(t)$ is $\mathcal{F}_t^M$ -measurable. Hence the martingale $X(t) = W^2(t) - t$ is adapted to $\mathcal{F}_t^M$, but is not an integral of a predictable process with respect to M . By Itô's formula

$X(t) = 2 \int_0^t W(s) dW(s)$. Hence $\langle X, M \rangle(t) = \int_0^t W^2(s) d\langle W, B \rangle(s) = 0$. Suppose there is H , such that $X(t) = \int_0^t H(u) dM(u)$. Then by (8.67)

$H(t) = \frac{d\langle X, M \rangle(t)}{dt} = 0$, implying that $X(t) = 0$, which is a contradiction.

This example has an application in Finance, it shows non-completeness of a stochastic volatility model.

Example 8.27: Let $\mathbb{F}$ be generated by a Brownian motions B and a Poisson process N , and let $M(t) = B(t) + N(t) - t = B(t) + \bar{N}(t)$, where $\bar{N}(t) = N(t) - t$. M is a martingale, as a sum of two martingales. We show that M does not have the predictable representation property.

$[M, M](t) = [B, B](t) + [N, N](t) = t + N(t)$. This shows that $N(t) = [M, M](t) - t$ is $\mathcal{F}_t^M$ -measurable. Hence $B(t) = M(t) - N(t) + t$ is $\mathcal{F}_t^M$ -measurable. Thus the martingale $X(t) = \int_0^t N(s-) dB(s)$ is $\mathcal{F}_t^M$ -measurable, but it does not have a predictable representation. If it did, $\int_0^t N(s-) dB(s) = \int_0^t H(s) dB(s) + \int_0^t H(s) d\bar{N}(s)$, and $\int_0^t (N(s-) - H(s)) dB(s) = \int_0^t H(s) d\bar{N}(s)$. Since the integral on the rhs is of finite variation, $H(s) = N(s-)$, for almost all s . Thus $\int_0^t H(s) d\bar{N}(s) = 0$. This is the same as $\int_0^t N(s-) dN(s) = \int_0^t N(s-) ds$. But this is impossible. To see the contradiction, let $t = T_2$ the time of the second jump of N . Then $\int_0^{T_2} N(s-) dN(s) = 1$ and $\int_0^{T_2} N(s-) ds = T_2 - T_1$.

This example shows non-completeness of models of stock prices with jumps. It can be generalized to a model that supports a martingale with a jump component.

Remark 8.7: The Definition 8.34 of the predictable representation property given here agrees with the standard definition given for continuous martingales

but is different to the definition for predictable representation with respect to semimartingales, given in Liptser and Shiryaev (1989) p.250, Jacod and Shiryaev (1987) p.172, Protter (1992) p.150. The general definition allows for different predictable functions h and H to be used in the integrals with respect to the continuous martingale part M^c and the discrete martingale part M^d of M ,

$$X(t) = X(0) + \int_0^t h(s) dM^c(s) + \int_0^t H(s) dM^d(s).$$

In this definition, the martingale M in Example 8.27 has the predictable representation property.

The definition given here is more suitable for financial applications. According to the financial mathematics theory an option can be priced if it can be replicated, which means that it is an integral of a predictable process H with respect to the discounted stock price process M , which is a martingale. The process H represents the number of shares bought/sold so it does not make sense to have H consist of two different components.

8.13 Elements of the General Theory

The basic setup consists of the probability space $(\Omega, \mathcal{F}, P)$, where $\mathcal{F}$ is a σ -field on Ω and P is a probability on $\mathcal{F}$. A stochastic process is a map from $\mathbb{R}^+ \times \Omega$ to $\mathbb{R}$, namely $(t, \omega) \rightarrow X(t, \omega)$. $\mathbb{R}^+$ has the Borel σ -field of measurable sets, and $\mathcal{F}$ is the σ -field of measurable sets on Ω . Only measurable processes are considered, that is for any $A \in \mathcal{B}$

$$\{(t, \omega) : X(t, \omega) \in A\} \in \mathcal{B}(\mathbb{R}^+) \times \mathcal{F}.$$

Theorem 8.40 (Fubini) *Let $X(t)$ be a measurable stochastic process. Then*

1. *P-a.s. the functions $X(t, \omega)$ (trajectories) are (Borel) measurable.*
2. *If $EX(t)$ exists for all t , then it is measurable as a function of t .*
3. *If $\int_a^b E|X(t)|dt < \infty$ P-a.s. then almost all trajectories $X(t)$ are integrable and $\int_a^b EX(t)dt = E \int_a^b X(t)dt$.*

Let $\mathbb{F}$ be a filtration of increasing σ -fields on Ω . Important classes of processes are introduced via measurability with respect to various σ -fields of subsets of $\mathbb{R}^+ \times \Omega$: adapted, progressively measurable processes, optional processes and predictable processes, given in the order of inclusion.

X is adapted if, for all t , $X(t)$ is $\mathcal{F}_t$ measurable. X is progressively measurable if, for any t , $\{(s \leq t, \omega) : X(s, \omega) \in A\} \in \mathcal{B}([0, t]) \times \mathcal{F}_t$. Any progressively measurable process is clearly adapted. It can be shown that any right or left continuous process is progressively measurable.

Definition 8.41

1. The σ -field generated by the adapted left-continuous processes is called the predictable σ -field $\mathcal{P}$.
2. The σ -field generated by the adapted right-continuous processes is called the optional σ -field $\mathcal{O}$.
3. A process is called predictable if it is measurable with respect to the predictable σ -field $\mathcal{P}$; it is called optional if it is measurable with respect to the optional σ -field $\mathcal{O}$.

Remarks

1. The predictable σ -field $\mathcal{P}$ is also generated by the adapted continuous processes.
2. Define $\mathcal{F}_{t-} = \sigma\left(\bigcup_{s < t} \mathcal{F}_s\right)$ the smallest σ -field containing $\mathcal{F}_s$ for all $s < t$, and $\mathcal{F}_{0-} = \mathcal{F}_0$. $\mathcal{F}_{t-}$ represents the information available prior to t . Then the predictable σ -field $\mathcal{P}$ is generated by the sets $[s, t) \times A$ with $s < t$ and $A \in \mathcal{F}_{t-}$.
3. Predictable and optional σ -fields are also generated respectively by simple adapted left-continuous processes and simple adapted right-continuous processes.
4. Since a left-continuous adapted process $H(t)$ can be approximated by right-continuous adapted processes, $(H(t) = \lim_{\epsilon \rightarrow 0} H((t - \epsilon)+))$, any predictable process is also optional. Therefore $\mathcal{P} \subseteq \mathcal{O}$.
5. The Poisson process is right-continuous and it can be shown that it cannot be approximated by left-continuous adapted processes. Therefore there are optional processes which are not predictable, $\mathcal{P} \subset \mathcal{O}$.
6. In discrete time optional is the same as adapted.

Stochastic Sets

Subset of $\mathbb{R}^+ \times \Omega$ are called stochastic sets. If A is a stochastic set, then “its projection on Ω ” $\pi_A = \{\omega : \exists t \text{ such that } (t, \omega) \in A\}$. A is called *evanescent* if $P(\pi_A) = 0$.

Two processes $X(t)$ and $Y(t)$ are called indistinguishable if the stochastic set $A = \{(t, \omega) : X(t, \omega) \neq Y(t, \omega)\}$ is an evanescent set. A process indistinguishable from zero is called P-negligible.

If τ_1 and τ_2 are stopping times, a closed stochastic interval is defined as $[[\tau_1, \tau_2]] = \{(t, \omega) : \tau_1(\omega) \leq t \leq \tau_2(\omega)\}$. Similarly half-closed $[[\tau_1, \tau_2[[$, $]]\tau_1, \tau_2]]$, and open $]]\tau_1, \tau_2[[$ stochastic intervals are defined. Double brackets are used to emphasize that the intervals are subsets of $\mathbb{R}^+ \times \Omega$ and to distinguish them from intervals on $\mathbb{R}^+$.

The stochastic interval $[[\tau, \tau]] = \{(t, \omega) : \tau(\omega) = t\}$ is called the graph of the stopping time τ .

A stochastic set A is called *thin* if there are stopping times τ_n , such that $A = \cup_n [[\tau_n]]$.

Example 8.28: Let $N(t)$ be a Poisson process with rate λ , and $A = \{\Delta N \neq 0\}$. Then $A = \cup_n [[\tau_n]]$, where τ_n is the time of the n -th jump. Hence A is a thin set.

It can be shown that for any regular right-continuous process the stochastic set of jumps $\{\Delta X \neq 0\}$ is a thin set (Liptser and Shiriyayev (1989), p.4).

Classification of Stopping Times

Recall that τ is a stopping time with respect to filtration $\mathbb{F}$ if for all $t \geq 0$ the event $\{\tau \leq t\} \in \mathcal{F}_t$. If $\mathbb{F}$ is right-continuous, then also $\{\tau < t\} \in \mathcal{F}_t$. Events observed before or at time τ are described by the σ -field $\mathcal{F}_\tau$, defined as the collection of sets $\{A \in \mathcal{F} : \text{for any } t, A \cap \{\tau \leq t\} \in \mathcal{F}_t\}$. Events observed before time τ are described by the σ -field $\mathcal{F}_{\tau-}$, the σ -field generated by $\mathcal{F}_0$ and the sets $A \cap \{\tau > t\}$, where $A \in \mathcal{F}_t$, $t > 0$.

There are three types of stopping times that are used in stochastic calculus:

1. predictable stopping times,
2. accessible stopping times,
3. totally inaccessible stopping times.

τ is a predictable stopping time if there exists a sequence of stopping times τ_n , $\tau_n < \tau$, and $\lim_n \tau_n = \tau$. In this case it is said that the sequence τ_n announces τ .

Example 8.29: If τ is a stopping time then for any constant $a > 0$, $\tau + a$ is a predictable stopping time. Indeed, it can be approached by $\tau_n = \tau + a - 1/n$.

Example 8.30: Let $B(t)$ be Brownian motion started at zero, $\mathbb{F}$ its natural filtration and τ the first hitting time of 1, that is, $\tau = \inf\{t : B(t) = 1\}$. τ is a predictable stopping time, since $\tau_n = \inf\{t : B(t) = 1 - 1/n\}$ converge to τ .

τ is an accessible stopping time if it is possible to announce τ , but with different sequences on different parts of Ω , that is, $[[\tau]] \subset \cup_n [[\tau_n]]$, where τ_n are predictable stopping times. All other types of stopping times are called totally inaccessible.

Example 8.31: Let $N(t)$ be Poisson process, $\mathbb{F}$ its natural filtration and τ the time of the first jump, $\tau = \inf\{t : N(t) = 1\}$. τ is a totally inaccessible stopping time. Any predictable stopping time $\tau_n < \tau$ is a constant, since $\mathcal{F}_t \cap \{t < \tau\}$ is trivial. But τ has a continuous distribution (exponential), thus it cannot be approached by constants.

The optional σ -field is generated by the stochastic intervals $[[0, \tau[$, where τ is a stopping time. The predictable σ -field is generated by the stochastic intervals $[[0, \tau]]$.

A set A is called predictable if its indicator is a predictable process, $I_A \in \mathcal{P}$.

The following results allow us to decide on predictability.

Theorem 8.42 *Let $X(t)$ be a predictable process and τ be a stopping time. Then*

1. $X(\tau)I(\tau < \infty)$ is $\mathcal{F}_{\tau-}$ measurable,
2. the stopped process $X(t \wedge \tau)$ is predictable.

For a proof see, for example, Liptser and Shiriyayev (1989) p. 13.

Theorem 8.43 *An adapted regular process is predictable if and only if for any predictable stopping time τ the random variable $X(\tau)I(\tau < \infty)$ is $\mathcal{F}_{\tau-}$ measurable and for each totally inaccessible stopping time τ one of the following two conditions hold*

1. $X(\tau) = X(\tau-)$ on $\tau < \infty$
2. the set $\{\Delta X \neq 0\} \cap [[\tau]]$ is $\mathbb{P}$ -evanescent.

For a proof see, for example, Liptser and Shiriyayev (1989) p. 16.

Theorem 8.44 *A stopping time τ is predictable if and only if for any bounded martingale M , $\mathbb{E}(M(\tau)I(\tau < \infty)) = \mathbb{E}(M(\tau-)I(\tau < \infty))$.*

Theorem 8.45 *The compensator $A(t)$ is continuous if and only if the jump times of the process $X(t)$ are totally inaccessible.*

See, for example, Liptser and Shiriyayev (2001) for the proof.

Example 8.32: The compensator of the Poisson process is t , which is continuous. By the above result the jump times of the Poisson process are totally inaccessible. This was shown in Example 8.31.

It can be shown that for Brownian motion and diffusions any stopping time is predictable. This implies that the class of optional processes is the same as the class of predictable processes.

Theorem 8.46 *For Brownian motion filtration any martingale (local martingale) is continuous and any positive stopping time is predictable. Any optional process is also predictable, $\mathcal{O} = \mathcal{P}$.*

Similar result holds for diffusions (see for example, Rogers and Williams (1990), p.338).

Remark 8.8: It can be shown that $\langle X, X \rangle$ is the conditional quadratic variation of $[X, X]$ conditioned on the predictable events $\mathcal{P}$.

8.14 Random Measures and Canonical Decomposition

The canonical decomposition of semimartingales with jumps uses the concepts of a random measure and its compensator, as well as integrals with respect to random measures. We do not use this material elsewhere in the book. However, the canonical decomposition is often met in research papers.

Random Measure for a Single Jump

Let ξ be a random variable. For a Borel set $A \subset \mathbb{R}$ define

$$\mu(\omega, A) = I_A(\xi(\omega)) = I(\xi(\omega) \in A). \quad (8.70)$$

Then μ is a random measure, meaning that for each $\omega \in \Omega$, $\mu(\omega, A)$ is a measure when A varies, $A \in \mathcal{B}(\mathbb{R})$. Its (random) distribution function has a single jump of size 1 at ξ . The following random Stieltjes integrals consist of a single term:

$$\int_{\mathbb{R}} x \mu(\omega, dx) = \xi(\omega), \text{ and for a function } h, \int_{\mathbb{R}} h(x) \mu(\omega, dx) = h(\xi(\omega)). \quad (8.71)$$

There is a special notation for this random integral

$$h * \mu := \int_{\mathbb{R}} h(x) \mu(dx). \quad (8.72)$$

Random Measure of Jumps and its Compensator in Discrete Time

Let $X_0, \dots, X_n, \dots$ be a sequence of random variables, adapted to $\mathcal{F}_n$, and let $\xi_n = \Delta X_n = X_n - X_{n-1}$. Let $\mu_n = I_A(\xi_n)$ be jump measures, and let

$$\nu_n(A) = E(\mu_n(A) | \mathcal{F}_{n-1}) = E(I_A(\xi_n) | \mathcal{F}_{n-1}) = P(\xi_n \in A | \mathcal{F}_{n-1})$$

be the conditional distributions, $n = 0, 1, \dots$. Define

$$\mu((0, n], A) = \sum_{i=1}^n \mu_i(A), \quad \text{and} \quad \nu((0, n], A) = \sum_{i=1}^n \nu_i(A). \quad (8.73)$$

Then for each A the sequence $\mu((0, n], A) - \nu((0, n], A)$ is a martingale. The measure $\mu((0, n], A)$ is called the measure of jumps of the sequence X_n , and $\nu((0, n], A)$ its compensator (A does not include 0). Clearly, the measure $\mu = \{\mu((0, n])\}_{n \geq 1}$ admits representation

$$\mu = \nu + (\mu - \nu), \quad (8.74)$$

where $\nu = \{\nu((0, n])\}_{n \geq 1}$ is predictable, and $\mu - \nu = \{\mu((0, n]) - \nu((0, n])\}_{n \geq 1}$ is a martingale. This is Doob's decomposition for random measures. With notation (8.72)

$$X_n = (x * \mu)_n. \quad (8.75)$$

Regular conditional distributions exist and for a function $h(x)$ the conditional expectations can be written as integrals with respect to these

$$E(h(\xi_n) | \mathcal{F}_{n-1}) = \int_{\mathbb{R}} h(x) \nu_n(dx),$$

provided $h(\xi_n)$ is integrable, $E|h(\xi_n)| < \infty$.

Assume now that ξ_n are integrable, then its Doob's decomposition (8.4)

$$X_n = X_0 + \sum_{i=1}^n E(\xi_i | \mathcal{F}_{i-1}) + \sum_{i=1}^n (\xi_i - E(\xi_i | \mathcal{F}_{i-1})) = X_0 + A_n + M_n. \quad (8.76)$$

Using random measures and their integrals, we can express A_n and M_n as

$$A_n = \sum_{i=1}^n E(\xi_i | \mathcal{F}_{i-1}) = \sum_{i=1}^n \int_{\mathbb{R}} x \nu_i(dx) = (x * \nu)_n, \quad (8.77)$$

$$M_n = \sum_{i=1}^n (\xi_i - E(\xi_i | \mathcal{F}_{i-1})) = \sum_{i=1}^n \int_{\mathbb{R}} x (\mu_i(dx) - \nu_i(dx)) = (x * (\mu - \nu))_n.$$

Thus the semimartingale decomposition of X is given by using the random measure and its compensator

$$X_n = X_0 + (x * \nu)_n + (x * (\mu - \nu))_n. \quad (8.78)$$

However, the jumps of X , $\xi_n = \Delta X_n$, may not be integrable. Then the term $(x * \nu)_n$ is not defined. In this case a truncation function is used, such as $h(x) = xI(|x| \leq 1)$, and a similar decomposition is achieved, called the canonical decomposition,

$$\begin{aligned} X_n &= X_0 + \sum_{i=1}^n E(h(\xi_i) | \mathcal{F}_{i-1}) + \sum_{i=1}^n (h(\xi_i) - E(\xi_i | \mathcal{F}_{i-1})) + \sum_{i=1}^n (\xi_i - h(\xi_i)) \\ &= X_0 + (h * \nu)_n + (h * (\mu - \nu))_n + ((x - h(x)) * \mu)_n. \end{aligned} \quad (8.79)$$

The above representation has well-defined terms and in addition it has another advantage that carries over to the continuous time case.

Random Measure of Jumps and its Compensator

Let X be a semimartingale. For a fixed t consider the jump $\Delta X(t)$. Taking $\xi = \Delta X(t)$, we obtain the measure of the jump at t

$$\mu(\{t\}, A) = I_A(\Delta X(t)), \quad \text{with} \quad \int_{\mathbb{R}} x \mu(\{t\}, dx) = \Delta X(t). \quad (8.80)$$

Now consider the measure of jumps of X (in $\mathbb{R}^+ \times \mathbb{R}^0$, with $\mathbb{R}^0 = \mathbb{R} \setminus 0$)

$$\mu((0, t] \times A) = \sum_{0 < s \leq t} I_A(\Delta X(s)) \quad (8.81)$$

for a Borel set A that does not include zero (there are no jumps of size 0, if $\Delta X(t) = 0$ then t is a point of continuity of X). It is possible to define the compensator of μ , such that $\mu((0, t] \times A) - \nu((0, t] \times A)$ is a local martingale.

For the canonical decomposition of semimartingales, similar to (8.79), firstly large jumps are taken out, then the small jumps are compensated as follows. Consider

$$(x - h(x)) * \mu(t) = \int_0^t \int_{\mathbb{R} \setminus 0} (x - h(x)) \mu(ds, dx) = \sum_{s \leq t} (\Delta X_s - h(\Delta X_s)),$$

where $h(x)$ is a truncation function. This is a sum over “large” jumps with $|\Delta X(s)| > 1$ (since $x - h(x) = 0$ for $|x| \leq 1$). Since the sum of squares of jumps is finite, (Corollary 8.31, (8.51)), the above sum has only finitely many terms, hence it is finite. Thus the following canonical decomposition of a semimartingale is obtained

$$\begin{aligned} X(t) &= X(0) + A(t) + X^{cm}(t) + (h(x) * (\mu - \nu))(t) + ((x - h(x)) * \mu)(t), \\ &= X(0) + A(t) + X^{cm}(t) + \int_0^t \int_{|x| \leq 1} x d(\mu - \nu) + \int_0^t \int_{|x| > 1} x d\mu, \end{aligned}$$

where A is a predictable process of finite variation, X^{cm} is a continuous martingale component of X , μ is the measure of jumps of X , and ν its compensator. For a proof see Liptser and Shiryaev (1989), p.188, Shiryaev (1999), p. 663.

Let $C = \langle X^{cm}, X^{cm} \rangle$ (it always exists for continuous processes). The following three processes appearing in the canonical decomposition (A, C, ν) are called the *triplet of predictable characteristics* of the semimartingale X .

Notes. Material for this chapter is based on Protter (1992), Rogers and Williams (1990), Metivier (1982), Liptser and Shiryaev (1989), Shiryaev (1999).

8.15 Exercises

Exercise 8.1: Let $\tau_1 < \tau_2$ be stopping times. Show that $I_{(\tau_1, \tau_2]}(t)$ is a simple predictable process.

Exercise 8.2: Let $H(t)$ be a regular adapted process, not necessarily left-continuous. Show that for any $\delta > 0$, $H(t - \delta)$ is predictable.

Exercise 8.3: Show that a continuous process is locally integrable. Show that a continuous local martingale is locally square integrable.

Exercise 8.4: M is a local martingale and $E \int_0^T H^2(s) d[M, M](s) < \infty$. Show that $\int_0^t H(s) dM(s)$ is a square integrable martingale.

Exercise 8.5: Find the variance of $\int_0^1 N(t-) dM(t)$, where M is the compensated Poisson process $M(t) - t$.

Exercise 8.6: If S and T are stopping times, show that

1. $S \wedge T$ and $S \vee T$ are stopping times.
2. The events $\{S = T\}$, $\{S \leq T\}$ and $\{S < T\}$ are in $\mathcal{F}_S$.
3. $\mathcal{F}_S \cap \{S \leq T\} \subset \mathcal{F}_T \cap \{S \leq T\}$.

Exercise 8.7: Let U be a positive random variable on a probability space $(\Omega, \mathcal{F}, P)$, and $\mathcal{G}$ be a sub- σ -field of $\mathcal{F}$.

1. Let $t \geq 0$. Show that $\mathcal{F}_t := \{A \in \mathcal{F} : \exists B \in \mathcal{G} \text{ such that } A \cap \{U > t\} = B \cap \{U > t\}\}$ is a σ -field.
2. Show that $\mathcal{F}_t$ is a right-continuous filtration on $(\Omega, \mathcal{F}, P)$.
3. Show that U is a stopping time for $\mathcal{F}_t$.

4. What are $\mathcal{F}_0$, $\mathcal{F}_{U-}$ and $\mathcal{F}_U$ equal to?

Exercise 8.8: Let $U_1, U_2, \dots$ be (strictly) positive random variables on a probability space $(\Omega, \mathcal{F}, P)$, and $\mathcal{G}_1, \mathcal{G}_2, \dots$ be sub- σ -fields of $\mathcal{F}$. Suppose that for all n , $U_1, U_2, \dots, U_n$ are $\mathcal{G}_n$ -measurable and denote by T_n the random variable $\sum_{i=1}^n U_i$. Set $\mathcal{F}_t = \bigcap_n \{A \in \mathcal{F} : \exists B_n \in \mathcal{G}_n \text{ such that } A \cap \{T_n > t\} = B_n \cap \{T_n > t\}\}$.

1. Show that $\mathcal{F}_t$ is a right-continuous filtration on $(\Omega, \mathcal{F}, P)$.

2. Show that for all n , T_n is a stopping time for $\mathcal{F}_t$.

3. Suppose that $\lim_n T_n = \infty$ a.s. Show that $\mathcal{F}_{T_n} = \mathcal{G}_{n+1}$ and $\mathcal{F}_{T_n-} = \mathcal{G}_n$.

Exercise 8.9: Let $B(t)$ be a Brownian motion and $H(t)$ be a predictable process. Show that $M(t) = \int_0^t H(s)dB(s)$ is a Brownian motion if and only if $\text{Leb}(\{t : |H(t)| \neq 1\}) = 0$ a.s.

Exercise 8.10: Let T be a stopping time. Show that the process $M(t) = 2B(t \wedge T) - B(t)$ obtained by reflecting $B(t)$ at time T , is a Brownian motion.

Exercise 8.11: Let $B(t)$ and $N(t)$ be respectively a Brownian motion and a Poisson process on the same space. Denote by $\bar{N}(t) = N(t) - t$ the compensated Poisson process. Show that the following processes are martingales: $B(t)\bar{N}(t)$, $\mathcal{E}(B)(t)\bar{N}(t)$ and $\mathcal{E}(\bar{N})(t)B(t)$.

Exercise 8.12: $X(t)$ solves the SDE $dX(t) = \mu X(t)dt + aX(t-)dN(t) + \sigma X(t)dB(t)$. Find the condition for $X(t)$ to be a martingale.

Exercise 8.13: Find the predictable representation for $Y = B^5(1)$.

Exercise 8.14: Find the predictable representation for the martingale $e^{B(t)-t/2}$.

Exercise 8.15: Let $Y = \int_0^1 \text{sign}(B(s))dB(s)$. Show that Y has a Normal distribution. Show that there is no deterministic function $H(s)$, such that $Y = \int_0^1 H(s)dB(s)$. This shows that the assumption that Y, B are jointly Gaussian in Theorem 8.35 is indispensable.

Exercise 8.16: Find the quadratic variation of $|B(t)|$.

Exercise 8.17: (Stochastic Logarithm)

Let U be a semimartingale such that $U(t)$ and $U(t-)$ are never zero. Show that there exists a unique semimartingale X with $X(0) = 0$, ($X = \mathcal{L}(U)$) such that $dX(t) = \frac{dU(t)}{U(t-)}$, and

$$X(t) = \ln \left| \frac{U(t)}{U(0)} \right| + \frac{1}{2} \int_0^t \frac{d \langle U^c, U^c \rangle(s)}{U^2(s-)} - \sum_{s \leq t} \left(\ln \left| \frac{U(s)}{U(s-)} \right| + 1 - \frac{U(s)}{U(s-)} \right), \quad (8.82)$$

see Kallsen and Shiryaev (2002).